

Re-evaluating Cosmic Composition: Non-Perturbative Averaging of Sub-100 Mpc Anisotropies as an Alternative to Dark Components

Input Question

What is the universe made of?

Domain

Astronomy

Validation Status

needs_data

Problem Statement

The composition of the universe includes measurable matter, dark matter, and dark energy. While standard cosmological models assign approximate proportions (5% measurable, 20-25% dark matter, 70-75% dark energy), the physical nature of dark components remains undefined ('dark spots'). Current evidence suggests these components may arise from averaging anisotropies on cosmological scales or indicate incompleteness in General Relativity, rather than representing distinct exotic substances.

Rationale

Understanding the universe's composition requires distinguishing between observable baryonic matter and inferred 'dark' components. Evidence indicates that homogeneity emerges only above ~ 100 Mpc, suggesting that sub-scale anisotropies might influence parameter inference. Furthermore, dark matter and energy are cited as hints of GR incompleteness. This research plan tests whether rigorous non-perturbative averaging of local structures can resolve these 'dark spots' without invoking new physics, thereby refining our understanding of what the universe is 'made of' in terms of effective gravitational sources.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Non-perturbative averaging of sub-100 Mpc structures in SDSS-IV eBOSS LRG and Planck 2018 data yields effective cosmological parameters deviating $>3\sigma$ from standard Λ CDM, potentially explaining 'dark spots' as backreaction artifacts.
- **Mechanism**: EV-Q053-5b6a5d5413f721935988346e establishes that homogeneity emerges only above ~ 100 Mpc after averaging anisotropies. If current FLRW-based analyses of SDSS-IV eBOSS and Planck 2018 data fail to fully capture non-linear backreaction from sub-100 Mpc structures, the inferred Ω_m and Ω_Λ may include systematic biases attributed to dark components. This hypothesis tests whether rigorous volume averaging

reduces these apparent exotic components.

- ****Falsifiable Prediction****: Applying non-perturbative backreaction formalisms (e.g., Buchert averaging) to SDSS-IV eBOSS LRG catalog and Planck 2018 CMB maps will produce effective $\Omega_{m,eff}$ and $\Omega_{\Lambda,eff}$ values that differ by $>3\sigma$ from Planck 2018 best-fit Λ CDM parameters AND provide a statistically superior fit ($\Delta\chi^2 < -10$) to Pantheon+ SNe Ia and DES Y3 BAO data compared to both standard Λ CDM and Green & Wald (2014) null-backreaction bounds.
- ****Required Observations****: SDSS-IV eBOSS LRG catalog (DR16) galaxy distribution at $z < 0.8$ for sub-100 Mpc structure reconstruction ; Planck 2018 CMB temperature and polarization maps for large-scale boundary conditions ; Pantheon+ Type Ia Supernovae sample for independent expansion history validation ; DES Y3 BAO measurements for geometric distance scale cross-check ; Computed effective density parameters under both standard perturbative and non-perturbative averaging schemes
- ****Risk of Being Wrong****: If non-perturbative averaging yields parameters consistent with Planck 2018 Λ CDM within 3σ , or fails to improve fit quality to Pantheon+/DES Y3 beyond Green & Wald (2014) limits, the hypothesis is falsified. Additionally, if local probes (e.g., SH0ES) remain inconsistent with averaged parameters, backreaction cannot resolve the tension.

Hypothesis 2

- ****Hypothesis****: Relic neutrino spectral distortions from early-universe interactions encode non-standard physics that correlates with late-time dark sector anomalies, offering a particle-physics explanation for cosmological 'dark spots'.
- ****Mechanism****: EV-Q053-1452363c14fbc7e57b667f1d documents that relic neutrino spectra are distorted by early-universe interactions. These distortions could carry imprints of dark sector couplings or sterile neutrinos that modify the effective number of relativistic species (N_{eff}) and matter-radiation equality epoch, thereby altering inferred dark matter/energy densities in ways not captured by standard Λ CDM.
- ****Falsifiable Prediction****: Theoretical modeling of neutrino spectral distortions under extended particle physics scenarios will predict N_{eff} and Σm_ν values that, when propagated through Boltzmann codes (CLASS/CAMB), produce CMB power spectra matching Planck 2018 residuals better than Λ CDM ($\Delta AIC > 6$) while remaining consistent with SDSS-IV eBOSS LRG clustering statistics.
- ****Required Observations****: Planck 2018 CMB TT/TE/EE power spectra and lensing reconstruction ; SDSS-IV eBOSS LRG correlation function measurements ; Numerical solutions of neutrino decoupling with non-standard interaction terms ; Derived constraints on N_{eff} and Σm_ν from joint CMB+LSS analysis
- ****Risk of Being Wrong****: If neutrino spectral distortion models cannot simultaneously fit Planck 2018 CMB and eBOSS LSS data better than Λ CDM, or if predicted N_{eff} values conflict with primordial nucleosynthesis bounds, this mechanism is excluded as an explanation for dark components.

Hypothesis 3

- ****Hypothesis****: Dark matter and dark energy phenomenology arises from infrared modifications to General Relativity manifesting at scales >100 Mpc, eliminating need for exotic substances.
- ****Mechanism****: EV-Q053-7e93b6230038a9c81916e34c identifies dark matter and dark energy as hints of GR incompleteness. Combined with EV-Q053-5b6a5d5413f721935988346e's statement that homogeneity emerges at

~100 Mpc, this suggests gravitational dynamics may transition to modified behavior precisely at the homogeneity scale, mimicking dark component effects without new particles.

- ****Falsifiable Prediction****: Modified gravity models calibrated to reproduce galactic rotation curves will predict lensing-galaxy cross-correlations in SDSS-IV eBOSS and weak lensing shear patterns in DES Y3 that deviate from Λ CDM predictions by $>5\%$ at $r > 10$ Mpc/h, while maintaining consistency with Solar System tests.
- ****Required Observations****: SDSS-IV eBOSS LRG-galaxy lensing cross-correlation measurements ; DES Y3 cosmic shear two-point correlation functions ; Galaxy rotation curve catalogs (SPARC/THINGS) ; Solar System ephemeris constraints on post-Newtonian parameters
- ****Risk of Being Wrong****: If lensing-galaxy correlations match Λ CDM within 5% at all scales, or if modified gravity models require fine-tuning to satisfy both cosmological and Solar System constraints, the hypothesis loses explanatory power relative to Λ CDM.

Technical Details

This experiment tests the hypothesis that non-perturbative averaging of sub-100 Mpc structures yields effective cosmological parameters deviating $>3\sigma$ from standard Λ CDM, potentially explaining 'dark spots' as backreaction artifacts. We will implement the Buchert averaging formalism to compute effective scale factors and curvature terms from inhomogeneous matter distributions, avoiding the assumption of global FLRW symmetry at small scales. The core technical challenge is reconstructing the local expansion history from SDSS-IV eBOSS LRG data and integrating it with Planck 2018 CMB boundary conditions. We will compare the inferred effective density parameters ($\Omega_{m,eff}$, $\Omega_{\Lambda,eff}$) against standard Λ CDM fits and the Green & Wald (2014) null-backreaction bounds. The analysis focuses on scales <100 Mpc where EV-Q053-5b6a5d5413f721935988346e notes anisotropies persist, testing if their proper averaging eliminates the need for exotic components or significantly alters their inferred magnitude.

Datasets

Source

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Target

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 "name": "DES Y3 BAO Measurements",
 "description": "Baryon Acoustic Oscillation data for geometric distance scale cross-checks.",
 "type": "validation_dataset"
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 "description": "Derived values of $\Omega_{m,eff}$ and $\Omega_{\Lambda,eff}$ under non-perturbative averaging schemes.",

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## Paper Abstract

Background: The standard cosmological model attributes ~95% of the universe's content to dark matter and dark energy, yet their physical nature remains unknown ('dark spots'). Evidence suggests that cosmic homogeneity emerges only above ~100 Mpc, implying that sub-scale anisotropies may influence global parameter inference. Methods: We propose a rigorous test of the hypothesis that non-perturbative averaging of sub-100 Mpc structures (using SDSS-IV eBOSS LRG data) yields effective cosmological parameters that deviate significantly from standard  $\Lambda$ CDM. We implement the Buchert averaging formalism and compare results against Planck 2018 CMB constraints and Green & Wald (2014) null-backreaction bounds. Validation Plan: Effective parameters ( $\Omega_{m,eff}$ ,  $\Omega_{\Lambda,eff}$ ) will be validated against independent probes: Pantheon+ SNe Ia and DES Y3 BAO data. Results: pending. This study aims to determine if 'dark' components are artifacts of incomplete averaging or require new physics.

## Methods

1. Data Preprocessing: Extract sub-100 Mpc anisotropy fields from SDSS-IV eBOSS LRG catalog (DR16) and align with Planck 2018 CMB maps. 2. Model Implementation: Implement the Buchert averaging formalism to compute effective scale factors, curvature, and backreaction terms from local inhomogeneities. 3. Null-Hypothesis Baseline: Compute backreaction effects using the Green & Wald (2014) rigorous bounds to establish a negligible-effect baseline. 4. Parameter Inference: Fit the effective expansion history derived from the averaged metric to Pantheon+ SNe Ia and DES Y3 BAO data. 5. Comparative Analysis: Calculate the deviation of  $\Omega_{m,eff}$  and  $\Omega_{\Lambda,eff}$  from Planck 2018 best-fit  $\Lambda$ CDM parameters and assess statistical significance ( $>3\sigma$  threshold).

## Experiments

### Baselines

```
```json
[
  "Standard  $\Lambda$ CDM Model: Assumes global homogeneity/isotropy and uses Planck 2018 best-fit parameters ( $\Omega_m \approx 0.315$ ,  $\Omega_\Lambda \approx 0.685$ ).",
  "Green & Wald (2014) Null-Backreaction Bounds: Rigorous theoretical limits demonstrating that backreaction effects are negligible in standard perturbation theory."
]
...
```

Metrics

```
```json
```

```
[
```

```
"Deviation in Ω_{dark} : Difference between inferred $\Omega_{\text{m,eff}}/\Omega_{\Lambda,\text{eff}}$ and Planck 2018 best-fit values, measured in standard deviations (σ).",
```

```
"Goodness-of-Fit ($\Delta \chi^2$): Improvement in fit quality to Pantheon+ SNe Ia and DES Y3 BAO data compared to standard Λ CDM (threshold: $\Delta \chi^2 < -10$).",
```

```
"Statistical Significance: Probability that the observed deviation from Λ CDM is due to random noise (target: $p < 0.003$ for $>3\sigma$)."
```

```
]
```

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```

Ablation

Vary the averaging scale threshold (50 Mpc, 100 Mpc, 200 Mpc) to test the specific claim from EV-Q053-5b6a5d5413f721935988346e that homogeneity emerges only above ~ 100 Mpc. Compare results with and without the Green & Wald correction terms.

Validation Protocol

Strictly validate the effective parameters against independent probes: Pantheon+ SNe Ia sample for expansion history and DES Y3 BAO measurements for geometric distances. If the refined averaging fails to match these independent datasets within error bars, or if the deviation from Planck 2018 is $<3\sigma$, the hypothesis is falsified.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q053-5b6a5d5413f721935988346e** · arxiv · arXiv:2409.07509
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2409.07509.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=9a4424d42c31df70724cbad05565651329f30223201579d0c83dc7829a4d406f
- **EV-Q053-7e93b6230038a9c81916e34c** · arxiv · arXiv:1711.08285
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1711.08285.pdf> · doi: Not available

• reliability_note: eligibility_status=FULLTEXT_VERIFIED;
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content_sha256=855942970bad322a20552d7b2324af4455812a1caa3297b11dddfafeb07f3705

• **EV-Q053-1452363c14fbc7e57b667f1d** · arxiv · arXiv:2210.10307

• authors: Not available · year: Not available

• url: <https://arxiv.org/pdf/2210.10307.pdf> · doi: Not available

• reliability_note: eligibility_status=FULLTEXT_VERIFIED;
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content_sha256=b969a8fec0bce91fba69008ff1d2bbb936be073bf54cdaecc9ba33ed2e7392

Reviewer Comments

- The revised hypothesis successfully addresses all critical issues from the previous review by specifying concrete datasets (SDSS-IV eBOSS LRG DR16, Planck 2018) instead of generic 'anisotropy maps'.
- Falsifiability is significantly improved with the inclusion of quantitative thresholds ($>3\sigma$ deviation, $\Delta\chi^2 < -10$) and a specific null-hypothesis baseline (Green & Wald 2014).
- Validation protocol now explicitly mandates independent probes (Pantheon+ SNe Ia, DES Y3 BAO) rather than leaving them conditional, satisfying reproducibility requirements.
- Evidence grounding remains strictly within allowed EvidenceCards; the link between 'dark spots' and averaging is correctly framed as a testable hypothesis derived from EV-Q053-5b6a5d5413f721935988346e rather than an established fact.
- Results field correctly states 'pending' with no fabrication of experimental outcomes.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Verify access to SDSS-IV eBOSS LRG catalog (DR16) and Planck 2018 CMB maps.
- Implement open-source code for Buchert averaging formalism.
- Include Green & Wald (2014) null-backreaction calculation as a baseline.
- Document the exact definition of 'effective' density parameters in the inhomogeneous context.
- Provide scripts for comparing refined vs. standard Λ CDM fits against Pantheon+ and DES Y3 data.